

EDUCATION

- **Sou Venutai Chavan Polytechnic** Pune, India
Bachelor of Technology - Information Technology Aug 2021 – May 2024
- **RMD Sinhgad Engineering College** Pune, India
Bachelor of Technology - Information Technology Aug 2024 – May 2027

SKILLS SUMMARY

- **Programming Languages:** Python, Java
- **Technologies:** GenAI, IoT, ESP32, Arduino
- **Core Subjects:** Data Structures, Algorithms, DBMS, OS
- **Databases:** MySQL
- **Software and Tools:** Jupyter, Visual Studio

EXPERIENCE

- **Detcher Tech Co.** Pune, India
Intern - Python Developer Jun 2023 – Jul 2023
 - **OTP Generator Development:** Designed and implemented a Python-based OTP generator using the random library to create secure 4-digit, 6-digit numeric, and 6-character alphanumeric codes.
 - **Enhanced Security Features:** Integrated logic to ensure randomness, uniqueness, and expiration for enhanced security in authentication processes.
 - **Optimization for Digital Services:** Optimized the system for use in online transactions, internet banking, and other digital services requiring secure access.
- **Incubators Systems Pvt. Ltd.** Pune, India
IoT & Robotics Intern Jan 2026 – Feb 2026
 - **IoT Development:** Developed IoT solutions using ESP32, Arduino, and Raspberry Pi for real-world applications.
 - **Cloud Integration:** Implemented IoT protocols and cloud-connected Streamlit dashboards for data monitoring.
 - **Robotics Experience:** Worked on robotics, simulation, and drone control systems to enhance automation skills.

PROJECTS

- **Smart Vehicle Black Box System:**
 - Developed an IoT-based vehicle black box using ESP32 for real-time accident monitoring.
 - Integrated multiple sensors to detect crashes, rollovers, and hazardous conditions.
 - Recorded critical vehicle data on an SD card for accident analysis and investigation.
 - Implemented emergency alerts and real-time monitoring to enhance road safety.
 - Built a Streamlit dashboard for data visualization and accident insights.
- **Smart Irrigation System:**
 - Designed and developed an Arduino-based smart irrigation system to automate plant watering.
 - Integrated soil moisture sensors to monitor moisture levels and send data to the Arduino.
 - Programmed the Arduino to trigger a relay-controlled water pump when the soil is dry.
 - Automated the system to turn off the pump after sufficient water is supplied, optimizing water usage.
 - Delivered a cost-effective solution that reduced manual effort and ensured consistent plant hydration.
- **Fire Alarm System:**
 - Designed and developed an automated fire detection and alarm system using an Arduino and flame sensor.
 - Programmed the flame sensor to detect flames and send a signal to the Arduino.
 - Configured the Arduino to activate a buzzer upon receiving the signal from the flame sensor.
 - Ensured timely response to fire hazards by triggering an alarm immediately when a flame is detected.

VOLUNTEER EXPERIENCE

- **Sinhgad Student Council** Pune, India
Assisted in organizing and managing events ensuring smooth execution. May 2022 – June 2023